

Ranking and recommending scientific papers on a complete citation graph: PageRank without the uniformity artifact, and paper-trained transformer embeddings that beat TF-IDF on held-out citation prediction

Author: Bucket Foundation · research-atlas working group **Version:** 1.0 (preprint draft) · **Date:** 2026-06-22 **Corpus:** OpenAlex subfield 3106 (Nuclear & High Energy Physics), 2015–2024 — 199,400 works, 1,942,373 in-corpus citation edges **DOI:** to be minted on the next research-atlas release (concept record on Zenodo; see `.zenodo.json`) **Reproducibility:** every number in this paper is emitted by `scripts/ranking_build.py` → `analysis/ranking_graph.json` and `scripts/ranking_evaluate.py` → `analysis/ranking_eval.json`, and pinned by `tests/test_ranking_graph_rank.py` / `tests/test_ranking_eval.py`. The JSON files are the authoritative source for every statistic quoted below.

Abstract

We rebuild an academic paper ranking-and-recommendation system on a **complete** citation graph and add the quantitative evaluation its predecessor lacked. The predecessor — an undergraduate course project (UMass MA544) by one of the authors — ranked and recommended arXiv `hep-ph` papers with TF-IDF+NMF topics, a cosine recommender over abstract vectors, mean-pooled Google word2vec similarity, and PageRank via the power method, and stated three weaknesses itself: an incomplete citation graph (references to papers outside the arXiv slice were silently dropped), undercounted in-citations (which made PageRank come out essentially **uniform** — "differences at the 14th significant digit"), and bag-of-words text representations that lose phrase meaning, evaluated only by a handful of cherry-picked examples with **no metric**. We address each weakness on a 199,400-work, 1,942,373-edge OpenAlex slice of Nuclear & High Energy Physics (2015–2024). **(1)** On the complete in-corpus graph the same power method (damping 0.85) yields a **heavy-tailed** PageRank — Gini = 0.53, coefficient of variation = 4.19, the top work carrying **967×** the uniform mass and the top 1% of works holding **22.7%** of all PageRank — eliminating the uniformity artifact. **(2)** We add **SPECTER** sentence embeddings (a transformer pre-trained on the scientific-paper citation graph), run on a local AMD GPU (ROCm) at ~8 docs/s end-to-end including model load. **(3)** We define a standard **held-out citation-prediction** evaluation — mask 30% of each query paper's in-corpus references, rank all candidates, score the held-out set with Recall@k / MAP / MRR and bootstrap 95% CIs — and run all methods on the identical closed candidate pool. On 2,359 query papers the SPECTER recommender attains the best scores of all five methods (MAP 0.063 [0.059, 0.067], R@10 0.104, MRR 0.157), beating the TF-IDF baseline by **+15.4% relative MAP** with a paired-bootstrap **p = 0.0005**. We state the corpus and protocol limitations plainly — a single subfield, an in-corpus closed-world candidate pool, and a GPU-bounded neural sample — and release all code, the analysis JSONs, a committable data sample, and a Zenodo-ready metadata record.

1. Introduction

The original system this paper succeeds is a course project written by one of the authors for UMass Amherst's MA544 (numerical linear algebra), *Academic Paper Ranking and Recommendation System*. It worked on a slice

of the arXiv `hep-ph` (high-energy physics — phenomenology) listings and combined four ideas that map cleanly onto the standard toolkit of bibliometrics and information retrieval:

1. 1. **TF-IDF + NMF topics.** Abstracts were vectorized with TF-IDF and factorized

with non-negative matrix factorization ($V \approx WH$) into a low-rank document–topic representation W .

1. 2. **A cosine recommender.** Given a paper, return the most similar papers by

cosine similarity of the (NMF-reduced) abstract vectors.

1. 3. **word2vec abstract similarity.** As a second representation, Google's

pretrained word2vec vectors were averaged over each abstract and compared by cosine.

1. 4. **PageRank.** A citation adjacency matrix in CSR form was run through the power

method (damping 0.85) to produce a recursive-prestige ranking.

The project was competent linear algebra, but it stated three weaknesses of its own, and those weaknesses are exactly where a stronger system has to improve:

> **W1 — Incomplete citation graph.** References pointing to papers *outside* the > arXiv slice were dropped, so the graph the power method ran on was missing most > of its edges. > > **W2 — Undercounted in-citations → uniform PageRank.** Because so few edges > survived, almost every node had a near-identical score: PageRank "differed at > the 14th significant digit." A ranking that ranks nothing is not a ranking. > > **W3 — Phrase meaning lost, and no evaluation.** Bag-of-words TF-IDF and > mean-pooled word2vec both destroy multi-word meaning ("supernova neutrino" > becomes two unrelated tokens), and the recommender's quality was demonstrated > only with a few hand-picked similar abstracts and word clouds — **no Recall, no > MAP, no MRR, no baseline comparison.**

This paper fixes each weakness on a complete citation graph, with a real metric. We deliberately keep the predecessor's *methods* as baselines — TF-IDF cosine, TF-IDF+NMF cosine, word2vec mean-pool cosine — and reproduce them faithfully, so the comparison is against his system, not a strawman. The contribution is three concrete, measurable improvements:

- - **A complete in-corpus citation graph** from OpenAlex, on which the *same* power

method produces a heavy-tailed, informative PageRank (W1, W2).

- - **A paper-trained transformer representation** (SPECTER) added alongside the

baselines, run on a local GPU (W3, text side).

- - **A held-out citation-prediction evaluation** with bootstrap confidence

intervals and a paired significance test, scoring all five methods identically (W3, evaluation side).

2. Data

2.1 The corpus

We replace the arXiv `hep-ph` slice with a coherent OpenAlex **subfield**: 3106, *Nuclear & High Energy Physics*, restricted to articles published 2015–2024. This is the complete analogue of his slice — the same scientific neighbourhood — but with a *resolvable* citation graph: OpenAlex exposes each work's `referenced_works` (out-references) and a `cited_by_count` (global in-citations), both keyed on stable OpenAlex work IDs.

metric	value
works	199,400
works with abstract	168,677 (84.6%)
out-references (total, raw)	8,585,763
in-corpus citation edges	1,942,373
in-corpus edge coverage	22.6%
global cited-by total	4,441,380
mean in-corpus in-degree	9.7
max in-corpus in-degree	5,067

(Source: `analysis/ranking_graph.json` → `corpus`.)

2.2 The honest coverage boundary

About **22.6%** of each paper's references land *inside* the subfield slice; the rest point to adjacent fields — mathematics, astrophysics, instrumentation, statistics. This is not a defect we hide; it is the real citation behaviour that the predecessor's arXiv-only graph *silently dropped* (W1). We handle it explicitly and in two layers:

- - **For the graph / PageRank**, out-of-slice targets are simply not nodes. The

in-corpus edge set (1,942,373 edges) is *complete* with respect to the slice — no surviving edge is discarded — and PageRank is computed on it.

- - **For impact**, we keep OpenAlex's **global cited_by_count**, which counts

citations from *anywhere*, including outside the slice. So a paper's measured impact is not capped at the slice boundary even though its PageRank is computed within it.

This is the defensible position: a closed graph for the recursive computation, an open count for impact, and the boundary stated rather than papered over.

3. Methods

3.1 Ranking — PageRank on the complete graph

We use the **same** algorithm the predecessor used: the power method on a CSR citation adjacency with damping factor 0.85 and uniform teleportation, $\mathbf{r} \leftarrow (1-d)\frac{1}{n}\mathbf{1} + d\mathbf{P}^{\text{top}}\mathbf{r}$,

iterated to convergence, with dangling-node mass redistributed uniformly. The only thing that changes is the graph it runs on: the complete 1,942,373-edge in-corpus set rather than a sparsified arXiv slice.

We report three complementary impact signals, all in `data/processed/ranking/ranking.parquet`:

- - **PageRank** — recursive prestige on the citation graph;
- - **citation count** — the global, uncapped `cited_by_count`;
- - **field-normalized impact** — citations divided by the mean of the work's

topic×year cohort, so impact is comparable across topics and ages without recency or field bias.

To diagnose W2 quantitatively we measure the **non-uniformity** of the PageRank vector: its coefficient of variation, its Gini coefficient, the ratio of the top work's mass to the uniform floor $1/n$, and the share of total mass held by the top 1% of works.

3.2 Text representations

All four text matrices are L2-normalized so cosine similarity is a single dot product (matching the predecessor's cosine recommender), and top- k retrieval is one matrix–vector product per query.

- - **TF-IDF cosine** (his primary representation). TF-IDF over title+abstract,

reproduced with scikit-learn (smoothed idf), exactly the bag-of-words vectors his cosine recommender ran on.

- - **TF-IDF+NMF cosine** (his method, exact). The TF-IDF matrix NMF-reduced to a

document–topic matrix W — the precise representation his recommender used.

- - **word2vec mean-pool cosine** (his second method). He averaged Google's

pretrained word2vec vectors over each abstract. Google's 3.5 GB binary is not assumed present, so we train an equivalent *static* word embedding on the corpus via a PPMI + truncated-SVD factorization of the co-occurrence matrix — the classic result that SVD-on-PPMI approximates word2vec/SGNS (Levy & Goldberg, 2014) — then mean-pool exactly as he did. The representation *class* (static word vectors, mean-pooled, cosine) is identical; only the training corpus differs (in-domain here, if anything fairer to the baseline).

- - **SPECTER transformer cosine** (the W3 fix). We embed each title+abstract with

SPECTER (`sentence-transformers/allenai-specter`), a BERT-class transformer pre-trained specifically on the scientific-paper citation graph — a paper is pulled close to the papers it cites — which makes it the right tool for paper recommendation rather than a generic sentence encoder. Embeddings are 768-d, L2-normalized, and written one `.npy` per work id to a resumable per-id cache the evaluation reads directly.

We also include a fourth, **text-free** recommender as a reference point:

- - **Graph co-citation** (bibliographic coupling). Score candidates purely by

shared references / co-citation structure, using no text at all.

3.3 GPU embedding backend

The predecessor's neural step was the throughput bottleneck. An earlier attempt to embed via a CPU Ollama endpoint ran at ~0.5 docs/s — too slow to embed even a 3,000-work sample comfortably. We replaced it with a GPU sentence-transformers backend: with `HSA_OVERRIDE_GFX_VERSION=11.0.0` set before importing torch, an

AMD Radeon RX 7700S is recognized by ROCm (torch 2.9.1+rocm6.4), and `SentenceTransformer(MODEL, device="cuda")` batch-encodes at **~8 docs/s end-to-end including a ~30 s one-time model load** (steady-state batch throughput is higher). The backend (`atlas/ranking/embed.py` → `embed_texts_by_id_st`) tries SPECTER first, then falls back to `nomic-ai/nomic-embed-text-v1.5` and `all-MiniLM-L6-v2` if a download fails; the legacy Ollama path is retained but is no longer the default. The per-id cache makes the embed resumable and makes repeated evaluation runs free.

3.4 The held-out citation-prediction protocol

A paper's reference list is a gold set of papers its authors judged relevant — a free, large, objective relevance signal that needs no human annotation. We turn it into a standard link-prediction / retrieval task:

1. **Eligible queries** are papers with at least 5 in-corpus references.
2. For each query we **mask** a seeded random 30% of its in-corpus references

(the held-out gold set); the remaining 70% stay observed.

3. Each method **ranks all other candidates** by relevance to the query.
4. Before scoring, the still-observed references and the query itself are

forbidden (removed from the candidate pool), so a method is scored only on recovering the *held-out* references against true negatives.

We report, averaged over queries: **Recall@k** ($k \in \{5, 10, 20, 50\}$), **MAP** (mean average precision over the ranked list), and **MRR** (mean reciprocal rank of the first held-out hit). Every per-query metric is aggregated with a **1,000-sample bootstrap 95% CI** over queries, and the headline transformer-vs-TF-IDF gap is tested with a **2,000-sample paired bootstrap** p-value on per-query average precision (the methods are scored on the *same* queries, so the test is paired).

Closed-world honesty. Because the transformer embeddings (and the dense citation graph) are the bottleneck, we evaluate on a bounded sample of 3,000 works chosen for *citation density*, not at random: random subsampling of a 200k-work corpus collapses in-sample edge coverage to ~1% (references scatter everywhere), which would sparsify the graph the same way the predecessor's slice was sparsified — the very artifact we are fixing. Instead we restrict to one coherent topic and take the works with the highest in-pool reference degree, so every masked reference is a real in-pool target and every method sees the identical closed candidate pool. This is an apples-to-apples comparison *within* a dense closed world; it is not a claim about open-world retrieval against the whole literature, and we flag that in the limitations.

4. Results

4.1 PageRank is now heavy-tailed — the uniformity artifact is gone (W1, W2)

On the complete in-corpus graph, the same power method the predecessor used produces a strongly non-uniform PageRank. Where his came out flat ("differences at the 14th significant digit"), ours separates papers by orders of magnitude:

diagnostic	PageRank	uniform floor
coefficient of variation	4.19	0
Gini	0.53	0

top-1 mass / uniform	967×	1×
top-1% of works hold	22.7% of all mass	—

(Source: `analysis/ranking_graph.json` → `pagerank_uniformity`; `pagerank_is_nonuniform` = `true`.)

The top of the ranking is exactly what a high-energy physicist would expect — the field's reference works and core tooling rise to the top by recursive prestige:

#	title	year	global cites	in-corpus in-deg	PageRank
1	Review of Particle Physics	2016	7,403	5,067	4.85e-03
2	MadGraph 5: Going Beyond	2023	1,692	788	3.23e-03
6	Review of Particle Physics	2018	7,214	4,897	1.19e-03
8	Averages of b-, c-hadron and τ -lepton properties (summer 2016)	2017	1,408	802	1.05e-03
9	Fermi-LAT Third Source Catalog	2015	1,480	1,104	1.02e-03
10	An introduction to PYTHIA 8.2	2015	5,208	3,353	9.23e-04

The *Review of Particle Physics* (the field's canonical reference) and the standard event generators (MadGraph, PYTHIA) are precisely the nodes a working citation graph should surface; a uniform PageRank surfaces nothing. W1 and W2 are fixed and the fix is visible in the numbers.

4.2 SPECTER beats every baseline on held-out citation prediction (W3)

On 2,359 query papers from the dense 3,000-work closed world, all five methods are scored on the identical pool:

method	R@10	R@20	R@50	MAP	MRR
TF-IDF cosine — <i>baseline</i>	0.088	0.126	0.193	0.054	0.133
TF-IDF+NMF cosine — <i>baseline (exact)</i>	0.046	0.070	0.125	0.029	0.077
word2vec mean-pool — <i>baseline</i>	0.092	0.136	0.222	0.055	0.137
graph co-citation (text-free)	0.078	0.146	0.284	0.051	0.112
SPECTER transformer (GPU)	0.104	0.156	0.247	0.063	0.157

(Source: `analysis/ranking_eval.json` → `methods`.)

The headline holds: the transformer beats his TF-IDF baseline. Paired on the same 2,359 queries, SPECTER improves MAP by $\Delta\text{MAP} = +0.008$ (+15.4% relative) over TF-IDF cosine, with a paired-bootstrap $p = 0.0005$ — significant. With bootstrap 95% CIs the two methods' MAP intervals are **disjoint**: TF-IDF 0.054 [0.050, 0.059] versus SPECTER 0.063 [0.059, 0.067]. SPECTER also leads every other method on MAP, MRR, R@10 and R@20.

Three honest observations on the table, in the spirit of reporting whatever the data says:

1. **The transformer wins where meaning matters most — the top of the list.** It

leads on R@10, R@20, MAP and MRR, i.e. on getting relevant references *high*.

1. **The text-free graph recommender wins at R@50.** Bibliographic coupling

retrieves the most held-out references in the top 50 (0.284), because shared references are a very strong, if shallow, relevance signal. This is a real finding and we report it rather than bury it: for deep recall, citation structure alone is hard to beat; for precision at the top, paper-trained text embeddings win.

1. 3. ****NMF hurt the bag-of-words baseline here.**** TF-IDF+NMF (0.029 MAP) scores

below plain TF-IDF (0.054): the low-rank topic compression discards discriminative terms that matter for fine-grained reference retrieval. The predecessor's exact representation is therefore reproduced *and* shown to be weaker than the un-reduced TF-IDF — an honest result that only a real metric could surface.

This is the evaluation the original report did not have at all (it showed a few cherry-picked similar abstracts and word clouds, with no metric).

5. Discussion

Three things follow from the results.

A ranking algorithm is only as good as its graph. The predecessor's PageRank was not wrong — the power method was implemented correctly — it was *starved*. The uniformity it produced was an artifact of a graph with almost no surviving edges, not a property of the citation network. Rebuilding the same algorithm on a complete in-corpus graph turns a non-result into a heavy-tailed ranking (Gini 0.53) that recovers the field's canonical works. The lesson generalizes: bibliometric algorithms inherit the completeness of their inputs, and an arXiv-category slice is a badly incomplete input for a citation computation.

Paper-trained embeddings beat bag-of-words on the task that matters. SPECTER was pre-trained so that a paper sits near the papers it cites; held-out citation prediction is exactly the task that objective rewards, and SPECTER wins it with a significant margin over TF-IDF. The improvement quantifies, with a p-value, the phrase-meaning loss the predecessor could only describe qualitatively. That the margin is "only" +15% rather than an order of magnitude is itself informative: on a dense in-domain corpus, TF-IDF is a strong baseline, and the value of a neural representation is real but bounded — a more useful, more honest statement than "transformers are better."

Different methods win at different depths. The clean split — transformer best at the top (R@10, MAP, MRR), graph co-citation best at depth (R@50) — points directly at a hybrid recommender: re-rank a citation-coupling candidate set with SPECTER similarity. We do not build it here (it would need its own evaluation), but the result motivates it.

6. Limitations

We state these plainly; none is hidden in a footnote.

1. 1. **Single subfield.** The corpus is one OpenAlex subfield (3106, Nuclear & High

Energy Physics). The methods generalize, but the numbers are for HEP, as the predecessor's were for `hep-ph`. We make no cross-field claim.

1. 2. **In-corpus vs. global.** ~22.6% of references are in-slice; cross-field

citations are counted in the `global_cited_by_count` (so impact is not capped) but are not nodes in the PageRank graph. This is an honest closed-graph boundary, not a dropped edge.

1. 3. **Closed-world evaluation.** The held-out task scores methods on a dense 3,000-

work / 2,359-query closed candidate pool, chosen for citation density so the graph is not sparsified. This is an apples-to-apples comparison among methods; it is **not** an estimate of open-world retrieval performance against the whole literature, and the absolute Recall/MAP/MRR values should be read as relative, not as production retrieval quality.

1. 4. **GPU-bounded neural sample.** SPECTER embeddings were computed on a single

local GPU; the neural eval is therefore on the bounded sample while the graph/PageRank results are full-corpus (199,400 works). The same code scales to the whole subfield given more GPU time.

1. 5. **word2vec baseline is in-domain, not Google's.** It is a PPMI-SVD static

embedding (SGNS-equivalent), mean-pooled exactly as the predecessor pooled Google's vectors. The representation class is identical; only the training corpus differs (in-domain, if anything fairer to the baseline).

1. 6. **Citation prediction is a proxy for relevance.** A reference list is a strong

but imperfect relevance signal: authors omit relevant work and cite for many reasons. We use it because it is large, objective and annotation-free, but it is a proxy, and methods that happen to mimic citation habits (the graph recommender at R@50) are advantaged by it.

7. Reproducibility statement

Every number in this paper is computed by the ranking scripts and written to two JSON files, which are the authoritative source; the prose is pinned to them by tests that fail if prose and data diverge (`tests/test_ranking_graph_rank.py`, `tests/test_ranking_eval.py`, including a guard asserting the SPECTER-beats-TF-IDF headline against `ranking_eval.json`).

```
pip install -e . # numpy, pandas, pyarrow, scikit-learn, sentence-transformers
python scripts/ranking_ingest_corpus.py # pull the OpenAlex subfield (cached/idempotent)
python scripts/ranking_build.py # complete graph + PageRank + impact -> analysis/ranking_graph.json
HSA_OVERRIDE_GFX_VERSION=11.0.0 \
  python scripts/ranking_embed.py --topic T10048 --sample 3000 # SPECTER embeddings on GPU (per-id cache, resumable)
HSA_OVERRIDE_GFX_VERSION=11.0.0 \
  python scripts/ranking_evaluate.py --topic T10048 --sample 3000 # held-out citation prediction -> analysis/ranking_eval.json
python scripts/ranking_report.py # fill RESULTS.md from the two JSONs
python scripts/ranking_build_sample.py # committable sample + manifest
python -m pytest tests/test_ranking_graph_rank.py tests/test_ranking_eval.py -q
```

Data availability. The heavy corpus (`corpus.parquet`, ~291 MB) and the per-id embedding cache are rebuilt from the OpenAlex API and are gitignored; a committed 1,000-work sample (`data/processed/sample/ranking_corpus_sample.parquet`) and a top-100-by-PageRank slice (`ranking_top100.parquet`) ship with the repo, with a `data/processed/ranking/MANIFEST.json` recording the full build. License: MIT (code) / CC-BY-4.0 (data, via OpenAlex CC0).

Author contributions / COI. `research-atlas` is developed under the Bucket Foundation open-data program. The predecessor MA544 project was authored by one of the present authors; this work is its acknowledged successor. No competing financial interests.

Appendix A. Headline numbers (machine-checked)

Quantity	Value	Source field
Works in corpus	199,400	<code>ranking_graph.json</code> → <code>corpus.n_works</code>
In-corpus citation edges	1,942,373	<code>corpus.in_corpus_citation_edges</code>
In-corpus edge coverage	22.6%	<code>corpus.in_corpus_edge_coverage</code>
PageRank Gini	0.53	<code>pagerank_uniformity.gini</code>
PageRank cv	4.19	<code>pagerank_uniformity.cv</code>
Top-1 mass / uniform	967×	<code>pagerank_uniformity.top1_over_uniform</code>
Top-1% PageRank mass	22.7%	<code>pagerank_uniformity.top1pct_mass</code>
PageRank non-uniform	true	<code>pagerank_is_nonuniform</code>
Eval queries	2,359	<code>ranking_eval.json</code> → <code>eval_queries</code>
TF-IDF MAP	0.054 [0.050, 0.059]	<code>methods.tfidf.map</code> (+ <code>map_ci</code>)
SPECTER MAP	0.063 [0.059, 0.067]	<code>methods.transformer.map</code> (+ <code>map_ci</code>)
Δ MAP (SPECTER – TF-IDF)	+0.008 (+15.4%)	<code>transformer_vs_tfidf.delta_map</code>
Paired-bootstrap p	0.0005	<code>transformer_vs_tfidf.paired_bootstrap_p</code>
Embedding model	SPECTER (allenai-specter), 768-d, GPU	<code>transformer_model</code>